

PROFESSIONAL SUMMARY

Software Developer and Mechatronics Engineer from RUET with strong multidisciplinary problem-solving foundations spanning enterprise backend architectures, scalable database modeling, modern web applications, and applied AI/ML systems. Experienced with C#, .NET, Java, Spring Boot, Python, and PostgreSQL, complemented by hands-on engineering competence in CAD solid modeling, CFD aerodynamic simulations, and physical prototyping. Passionate about building robust, maintainable software systems, intelligent LLM-assisted applications, and practical technology products.

TECHNICAL SKILLS

Programming:	C#, Java, Python, JavaScript, TypeScript, SQL	Backend & Web:	.NET, ASP.NET, ASP.NET MVC, Spring Boot, REST APIs, EF Core
Databases:	PostgreSQL, SQL, Relational Schema Design	Frontend:	React, HTML5, CSS3, JavaScript, TypeScript, Bootstrap, Tailwind CSS
AI & LLMs:	Machine Learning, NLP, Large Language Models, AI Agents, Tool Calling	Engineering:	CAD (Fusion 360), CFD (ANSYS CFX), FEA, Robotics, Automation, 3D Printing

KEY PROJECTS

Vertical Axis Wind Turbine for Low Wind Speed Area

Undergraduate Thesis & Experimental Prototype

RUET Mechatronics Lab

Autodesk Fusion 360 • ANSYS CFX • CFD Simulation • NACA 0021 Airfoil • 3D Printing

- Designed and investigated a compact three-bladed H-Darrieus vertical axis wind turbine tailored for low-wind-speed urban environments (Rotor: Ø 650 mm x H 1000 mm, Design Wind Speed: 5 m/s, Investigated Angular Velocity: 9 rad/s).
- Constructed parametric 3D CAD models of the central mast, support struts, and hollow aerodynamic blades using Fusion 360.
- Conducted Computational Fluid Dynamics (CFD) aerodynamic simulations in ANSYS CFX to compute torque coefficients, pressure distributions, and boundary-layer separation characteristics across varying angles of attack.
- Fabricated physical prototype blade sections via precision additive manufacturing (3D printing) to evaluate rotational balance and low-wind self-start response.

Financial AI Platform

Product Concept

Independent Development

Python • LLM APIs • Document Processing • Excel Automation • PowerPoint Generation • Retrieval

- Architected an AI productivity platform concept automating financial modeling, corporate filing extraction, spreadsheet formulation, and presentation synthesis.
- Designed retrieval pipelines linking unstructured financial disclosures with contextual LLM reasoning for accurate metric computation.
- Mapped programmatic execution flows generating formulaic Excel spreadsheets and executive decks directly from synthesized financial data.

E-commerce Platform

Web Application Concept

Full-Stack Development

C# / ASP.NET or Spring Boot • PostgreSQL • RESTful APIs • React • bKash Payment Workflow

- Designed a lightweight, responsive e-commerce web application architecture tailored for regional commerce with bilingual (Bangla/English) support.
- Engineered relational data models and RESTful backend endpoints in PostgreSQL for inventory tracking, catalog search, and administrative order management.
- Integrated localized checkout logic supporting both Cash on Delivery (COD) and bKash digital payment verification workflows.

- Explored multi-channel communication automation across WhatsApp Business, Facebook, and Instagram to reduce response latency for online retailers.
- Designed intent-detection pipelines with AI agents for automated FAQ resolution, order inquiry triaging, and seamless operator escalation.

TECHNICAL EXPERIENCE & DEVELOPMENT

Software Engineering & Backend Systems Focus

Active Development

- Designing scalable server architectures, REST APIs, and database migrations with C#, .NET, ASP.NET Core, and Entity Framework Core.
- Deepening full-stack competencies with Java, Spring Boot, relational SQL databases (PostgreSQL), and modern frontends with React and TypeScript.
- Practicing disciplined algorithms, Data Structures (DSA), version control workflows (Git/GitHub), and clean code standards.

Applied AI, NLP & Intelligent Agent Engineering

Independent Research

- Constructing structured agentic workflows, prompt engineering frameworks, and domain-context retrieval pipelines utilizing modern LLM APIs.
- Experimenting with NLP text extraction, financial document parsing, and practical task automation tools.

EDUCATION

Bachelor of Science in Mechatronics Engineering

Rajshahi University of Engineering & Technology (RUET)

Department of Mechatronics Engineering • Rajshahi, Bangladesh

Core Coursework: Control Systems Engineering, Robotics & Automation, Fluid Mechanics & Aerodynamics, CAD/CAM, Microprocessors & Microcontrollers, Finite Element Analysis (FEA), Engineering Mechanics & Dynamics, Data Structures & Computer Programming, Industrial Instrumentation & Sensors.

TECHNICAL INTERESTS

- Software Engineering & Backend Architecture • Artificial Intelligence & LLM Applications
- Robotics & Industrial Automation • Product Development • Technology Entrepreneurship